

Four-Corner Spectral Tuning for Quadratic ADMM with Application to Diffeomorphic Image Registration

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Abstract

Selecting the penalty and relaxation parameters of the alternating direction method of multipliers (ADMM) can require repeated spectral-radius evaluation of a large parameter-dependent iteration operator. We consider over-relaxed ADMM (oADMM) for a separable quadratic split whose attained joint modal spectrum is a Cartesian product of data and regularizer curvatures. The modal contraction factor is monotone along each curvature coordinate line; consequently, its largest magnitude is attained among four endpoint combinations. The relaxation parameter is optimized analytically for each penalty, and the remaining penalty problem reduces to a finite algebraically characterized superset of candidate minimizers. In this sense the method is search-free: it avoids grid-based or iterative numerical optimization over the penalty while still computing the required polynomial roots numerically. Full diffeomorphic image registration is spatially varying and noncommuting, so the exact theorem does not apply to the assembled operator. Coefficient freezing and the rank-one optical-flow Hessian nevertheless yield an inexpensive predictor determined by the four combinations of h_- , h_+ , g_- , and g_+ . On 134 FIRE retinal pairs at 256×256 , one-shot tuning reduced median pairwise runtime by 34.4% and inner iterations by 39.6% relative to a fixed pair selected on separate validation data. On ten CIMA histology pairs, runtime decreased by 20.1%. In both studies, landmark accuracy was essentially unchanged and no nonpositive discrete Jacobian was observed.

Keywords: alternating direction method of multipliers; parameter selection; spectral analysis; diffeomorphic image registration; local Fourier analysis

1 Introduction

Spectral parameter selection for quadratic ADMM can be expensive in its own right. For a trial penalty ρ , a spectral method constructs or applies the corresponding iteration operator, estimates its convergence factor, and repeats this calculation while optimizing ρ ; over-relaxation introduces the additional parameter α . In large imaging systems the operator may have hundreds of thousands of unknowns, so repeated spectral-radius evaluation can offset the computational advantage of splitting. This issue is especially pronounced after spatially varying image terms are assembled and whenever relinearization changes the local Hessian.

The structural question addressed here is whether this spectral optimization can collapse analytically when the quadratic operators possess separable modal structure. For the class studied below, the oADMM contraction of a common mode depends only on a data curvature h and a regularizer curvature g . For fixed ρ , this scalar factor is monotone along each coordinate line,

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with the direction determined by the other curvature. If the attained joint modal spectrum is the Cartesian product of the two curvature sets, the worst contraction must therefore occur among

$$(h_-, g_-), \quad (h_-, g_+), \quad (h_+, g_-), \quad (h_+, g_+).$$

The large spectral optimization is reduced exactly to four rational scalar branches. The optimal relaxation is available in closed form for each penalty, and the reduced penalty objective is piecewise rational. Its global minimum on a prescribed compact interval belongs to a finite algebraically characterized candidate set.

We call this parameter selection *search-free*: it performs neither a grid search nor an iterative numerical optimization over ρ . Polynomial roots in the finite candidate construction are still evaluated numerically; the distinction is the replacement of repeated optimization and spectral-radius evaluation by a fixed algebraic reduction.

The exact result requires more than commutativity. Simultaneous diagonalization supplies paired eigenvalues (h_i, g_i) , but these pairs need not fill the Cartesian product of the marginal spectra. The four endpoint combinations are exact modes only under the separability and attainment assumptions stated in Theorem 4.2. For arbitrary commuting operators they define a rectangular envelope, and for noncommuting operators they define a surrogate unless an independent bound is available.

Diffeomorphic image registration provides the motivating noncommuting application. Its linearized data Hessian varies by pixel, whereas the Sobolev regularizer couples neighboring pixels; the assembled matrices therefore generally do not commute. Freezing the image coefficient recovers the separable model locally. Moreover, each optical-flow Hessian is a rank-one update of a multiple of the identity, so all required data curvature information reduces to

$$h_- = \nu, \quad h_+ = \nu + \mu \max_i \|\nabla I(x_i)\|^2.$$

Together with the regularizer endpoints g_- and g_+ , these values yield a four-branch predictor after one $O(n)$ scan of gradient magnitudes. The result is exact for the frozen separable family and predictive for the full variable-coefficient registration operator.

The mathematical and computational progression is

$$\begin{aligned} \text{quadratic ADMM} &\longrightarrow \text{modal factor} \longrightarrow \text{four corners} \\ &\longrightarrow \text{finite parameter selection} \longrightarrow \text{noncommuting registration.} \end{aligned}$$

The contributions are:

1. A four-corner spectral theorem for quadratic oADMM under an attained Cartesian-product joint modal spectrum.
2. A finite algebraic joint selection of ρ and α that eliminates grid-based or iterative penalty search for this structured model.
3. A registration specialization based on local coefficient freezing and the rank-one optical-flow Hessian, reducing the data-term information required for tuning to endpoint curvatures.
4. Experimental evidence that the surrogate-optimal parameters transfer to noncommuting registration systems, reducing iterations and runtime while maintaining landmark accuracy and observed topology.

2 Related Work

2.1 ADMM parameter selection

Scaled ADMM and its practical residual criteria were systematized by Boyd et al. [2]. Ghadimi et al. derived optimal parameters for specific quadratic classes [5]. Teixeira et al. connected parameter selection with scaling and constraint preconditioning in distributed quadratic programs [13], while Giselsson and Boyd developed tight contraction bounds that depend on the chosen metric [6]. These works obtain strong conclusions from particular quadratic structure or operator bounds. The present analysis instead identifies a joint modal structure under which the full worst-mode calculation reduces to four attained endpoint pairs.

Adaptive methods avoid an offline spectral model. Residual balancing changes the penalty according to primal and dual residual magnitudes [17]; Barzilai–Borwein (BB) spectral updates estimate local curvature from iterate differences [18]. Such rules apply beyond the separable setting considered here, but they update parameters from the observed iteration history rather than solving the fixed quadratic model’s spectral minimax problem algebraically.

2.2 Spectral and over-relaxed ADMM

Song et al. address a broader class of linear-quadratic ADMM problems and numerically optimize the spectral convergence factor with respect to the penalty [12]. For each candidate ρ , their formulation requires spectral information from the corresponding iteration operator; the relaxation parameter is then selected conditionally. This generality is valuable because it does not require the Cartesian-product modal structure used here.

Direct application to full-resolution registration is computationally demanding: the assembled variable-coefficient operator is large and noncommuting, and its spectrum changes when image warping changes the linearized Hessian. Repeated spectral-radius evaluation across trial penalties can therefore make parameter selection expensive. A downsampled operator reduces this cost but introduces a transfer problem between the coarse spectral model and the full-resolution solve. The present work adopts the complementary tradeoff. Stronger separability assumptions collapse the structured spectral problem to four scalar branches and a finite penalty candidate set; the resulting construction is then used as an inexpensive predictor outside that exact class. A coarse direct-spectral diagnostic illustrating this overhead and transfer issue is reported in the supplementary information.

2.3 Fourier and local Fourier analysis

Fourier analysis has long characterized stationary iterations for elliptic problems [4], and local Fourier analysis (LFA) is a standard tool for multigrid design and parameter optimization [16]. Translation-invariant periodic problems admit exact Fourier decompositions; local and nonstandard analyses can remain informative for random or discontinuous coefficients [11, 9]. Here the stationary map is an ADMM error iteration, and local freezing converts a matrix-valued optical-flow coefficient into a separable patch model.

2.4 Variational image registration

DARTEL [1], diffeomorphic demons [15], and large-deformation optimal-control formulations [10] are representative optimization-based registration methods. Thorley et al. compose nonstationary velocity updates and solve regularized convex subproblems with accelerated ADMM on graphics

processing units (GPUs) [14]. Song et al. use a related quadratic model in their spectral oADMM study [12]. Learning-based methods broaden the current landscape [7], but the present work concerns numerical parameter selection for the linearized variational subproblem.

3 Quadratic ADMM and Reduced Error Dynamics

3.1 Quadratic splitting

Consider

$$\min_{v,w} \underbrace{\frac{1}{2}v^\top H v + b^\top v}_{f(v)} + \underbrace{\frac{1}{2}w^\top G w}_{r(w)} \quad \text{subject to} \quad v - w = 0, \quad (1)$$

where $v, w, b \in \mathbb{R}^N$ and $H, G \in \mathbb{R}^{N \times N}$ are symmetric positive semidefinite matrices. The linear term b determines the fixed point but disappears from homogeneous error propagation.

3.2 Standard and over-relaxed ADMM

With scaled dual variable u and penalty $\rho > 0$, standard ADMM is

$$(H + \rho I)v^{k+1} = -b + \rho(w^k - u^k), \quad (2)$$

$$(G + \rho I)w^{k+1} = \rho(v^{k+1} + u^k), \quad (3)$$

$$u^{k+1} = u^k + v^{k+1} - w^{k+1}. \quad (4)$$

For over-relaxed ADMM, abbreviated *oADMM*, define

$$\hat{v}^{k+1} = \alpha v^{k+1} + (1 - \alpha)w^k, \quad 0 < \alpha \leq 2, \quad (5)$$

and replace v^{k+1} by $\hat{v}^{k+1}$ in (3) and the dual update.

3.3 Reduced Cayley representation

The original oADMM recurrence couples primal and dual errors. Reducing it to one N -dimensional operator isolates all dependence on ρ and α and exposes the common-mode scalar factor used in the four-corner argument. Define

$$C_H(\rho) = (H - \rho I)(H + \rho I)^{-1}, \quad C_G(\rho) = (G - \rho I)(G + \rho I)^{-1}.$$

Proposition 3.1 (Reduced Cayley error operator). *The nonzero spectra of the oADMM state map and*

$$E_{\rho,\alpha} = \left(1 - \frac{\alpha}{2}\right) I + \frac{\alpha}{2} C_H(\rho) C_G(\rho). \quad (6)$$

coincide, including algebraic multiplicity; the remaining state-map eigenvalues are zero. Thus the asymptotic factor is $\varrho(E_{\rho,\alpha})$, where $\varrho(\cdot)$ denotes the spectral radius of a matrix or linear operator.

Appendix A gives the reduction. It is supporting machinery: the parameter reduction below comes from the modal dependence of (6).

Theorem 3.2 (Basic contraction). *If $H, G \succeq 0$ and $H + G \succ 0$, then for every $\rho > 0$, $\kappa_\rho := \|C_H(\rho)C_G(\rho)\|_2 < 1$. Consequently*

$$\varrho(E_{\rho,\alpha}) \leq 1 - \frac{\alpha}{2}(1 - \kappa_\rho) < 1,$$

for $0 < \alpha \leq 2$.

Proof. For $A \succeq 0$, orthogonal diagonalization shows $\|C_A(\rho)\|_2 \leq 1$, with norm preservation possible only on $\ker A$. If a unit vector x satisfied $\|C_H C_G x\|_2 = 1$, equality throughout $\|C_H C_G x\|_2 \leq \|C_G x\|_2 \leq 1$ would give $x \in \ker G$ and $C_G x = -x \in \ker H$. Hence $x \in \ker H \cap \ker G$, contradicting $H + G \succ 0$. The continuous function $x \mapsto \|C_H C_G x\|_2$ attains its maximum on the finite-dimensional unit sphere, so $\kappa_\rho < 1$. Finally, the triangle inequality applied to (6) gives the stated bound. $\square$

Nullspaces. If a nonzero vector lies in $\ker H \cap \ker G$, then $C_H C_G x = x$ and $E_{\rho, \alpha} x = x$ for every admissible parameter pair. Strict contraction therefore requires $H + G \succ 0$, or restriction to the identifiable subspace. In the four-corner registration model, $\nu = \gamma = 0$ creates the $(h, g) = (0, 0)$ mode; the selector then requires positive damping, a coercive boundary/regularization condition, or explicit removal of that common nullspace.

3.4 Scalar modal factor

When a vector is a common eigenvector of H and G , with eigenvalues h and g , (6) acts by

$$e(h, g; \rho, \alpha) = 1 - \frac{\alpha}{2} + \frac{\alpha}{2} \frac{h - \rho}{h + \rho} \frac{g - \rho}{g + \rho} \quad (7)$$

$$= 1 - \alpha + \alpha \theta(h, g; \rho),$$

$$\theta(h, g; \rho) = \frac{\rho^2 + hg}{(\rho + h)(\rho + g)}. \quad (8)$$

For nonnegative h and g , $0 \leq \theta \leq 1$. The paired values (h, g) , not the marginal spectra separately, determine the true modal factor.

4 Four-Corner Spectral Tuning

4.1 Separable commuting modal structure

Definition 4.1 (Cartesian-product joint modal spectrum). A quadratic split is *separable* on modal sets $\mathcal{H}$ and $\mathcal{G}$ if H and G are simultaneously diagonalizable and their attained joint modal spectrum is

$$\{(h, g) : Hx = hx, Gx = gx \text{ for a common mode } x \neq 0\} = \mathcal{H} \times \mathcal{G}.$$

For example, $H_{\text{sep}} = I_m \otimes H_c$ and $G_{\text{sep}} = G_s \otimes I_d$ have modes $\phi_\xi \otimes z_j$ and pairs $(h_j, g(\xi))$, where $H_c z_j = h_j z_j$, $G_s \phi_\xi = g(\xi) \phi_\xi$, $H_c \in \mathbb{R}^{d \times d}$ acts on components, $G_s \in \mathbb{R}^{m \times m}$ acts on spatial degrees of freedom, and I_m, I_d are the corresponding identity matrices.

The Cartesian-product condition is stronger than commutativity. It asserts that data and regularizer modes combine independently.

4.2 Four-corner theorem

For fixed $\rho > 0$,

$$\frac{\partial \theta}{\partial h} = \frac{\rho(g - \rho)}{(g + \rho)(h + \rho)^2}, \quad \frac{\partial \theta}{\partial g} = \frac{\rho(h - \rho)}{(h + \rho)(g + \rho)^2}. \quad (9)$$

The sign of the first derivative is independent of h , and the sign of the second is independent of g . Thus, for fixed ρ , θ is monotone along each coordinate line, although the direction may depend on the other coordinate. Its minimum and maximum are therefore attained at corners. Relaxation

maps θ affinely to $1 - \alpha + \alpha\theta$; taking absolute value produces a convex function of θ , whose maximum over an interval is again attained at an endpoint.

Theorem 4.2 (Four-Corner Spectral Theorem). *Suppose the joint modal spectrum is the Cartesian product $\mathcal{H} \times \mathcal{G}$, where $\mathcal{H} \subset [h_-, h_+]$ and $\mathcal{G} \subset [g_-, g_+]$, with $0 \leq h_- \leq h_+$ and $0 \leq g_- \leq g_+$. If all four endpoint combinations are attained, then*

$$\varrho(E_{\rho, \alpha}) = U_{4C}(\rho, \alpha) := \max_{\substack{h \in \{h_-, h_+\} \\ g \in \{g_-, g_+\}}} |e(h, g; \rho, \alpha)| \quad (10)$$

for every $\rho > 0$ and $0 < \alpha \leq 2$. If only the interval enclosure is known, the right-hand side remains a valid rectangular upper envelope, but equality with the true spectral radius need not hold.

Proof. Equation (9) makes the extrema of θ on the rectangle occur at corners. The map $t \mapsto |1 - \alpha + \alpha t|$ is convex, so its maximum over the interval between those extrema also occurs at an endpoint and hence at a corner. Cartesian-product structure and attained endpoint combinations make those corners actual modes, proving equality. $\square$

4.3 Optimal relaxation for fixed penalty

Let the four corner values satisfy

$$a(\rho) = \min_c \theta_c(\rho), \quad b(\rho) = \max_c \theta_c(\rho).$$

The four-corner objective is

$$\max\{|1 - \alpha + \alpha a|, |1 - \alpha + \alpha b|\}.$$

When $b < 1$, equioscillation of the two endpoint factors gives the unique unconstrained optimizer; accounting for $0 < \alpha \leq 2$ yields

$$\alpha^*(\rho) = \min \left\{ 2, \frac{2}{2 - a(\rho) - b(\rho)} \right\}. \quad (11)$$

After eliminating α , the reduced objective is

$$\begin{aligned} \widehat{U}(\rho) &= \frac{b - a}{2 - a - b} && \text{if } a + b \leq 1, \\ \widehat{U}(\rho) &= 2b - 1 && \text{if } a + b \geq 1. \end{aligned} \quad (12)$$

If $b = 1$, the objective equals one for every admissible α ; any relaxation may be selected, but the corresponding corner mode or envelope is not contractive. Appendix C gives the constrained equioscillation argument.

4.4 Finite algebraic penalty selection

After eliminating α , the objective is piecewise rational in ρ . Its expression can change only when the corner attaining $a(\rho)$ changes, the corner attaining $b(\rho)$ changes, or the relaxation regime crosses $a(\rho) + b(\rho) = 1$. Between these events, the reduced objective is one smooth rational function. A minimum on each closed piece must therefore occur at a piece endpoint or a stationary point. This converts the continuous penalty problem into finitely many algebraic conditions.

Formally, first deduplicate identical rational corner branches $\theta_c(\rho)$. On a compact admissible interval $\mathcal{I} = [\rho_-, \rho_+] \subset (0, \infty)$, changes in the active minimum or maximum occur only at pairwise branch intersections.

Theorem 4.3 (Finite penalty candidate set). *A global minimizer of (12) over $\mathcal{I}$ is contained in the finite set formed by:*

1. ρ_- and ρ_+ ;
2. positive pairwise branch-intersection roots in $\mathcal{I}$;
3. roots of $a(\rho) + b(\rho) = 1$ for every active branch pair; and
4. stationary roots of each active rational expression in (12).

Identically zero intersection, transition, or stationarity polynomials are discarded; a constant objective piece is represented by its interval endpoints. Thus the candidate set remains finite when endpoints coincide. After positive denominators are cleared, the respective polynomial degrees are at most three, four, and six. Evaluating (11) and (10) on this set—using any admissible α when $b = 1$ —returns a global minimizer of the four-corner objective over $\mathcal{I} \times (0, 2]$.

The proof is in Appendix D. The listed roots form a finite *superset*: roots outside $\mathcal{I}$, roots associated with inactive branches, and extraneous roots introduced by denominator clearing are filtered by evaluating the original objective. Search-free selection refers to this finite algebraic evaluation in place of grid-based or iterative optimization over ρ ; numerical polynomial root finding is still required.

4.5 Arbitrary commuting operators versus Cartesian products

Three cases must be distinguished.

Case A: For a separable commuting split with attained endpoint combinations, (10) is the true spectral radius, and finite tuning is globally optimal for that model over $\mathcal{I} \times (0, 2]$.

Case B: If $HG = GH$ but the paired spectrum is only $\{(h_i, g_i)\}_i$, the exact factor is $\max_i |e(h_i, g_i; \rho, \alpha)|$. The four corners of the marginal spectral rectangle can be artificial and provide an envelope rather than an equality.

Case C: If $HG \neq GH$, no global scalar modal decomposition follows. Four-corner tuning is a surrogate or predictor unless a separate argument supplies a bound.

5 Spatially Varying and Noncommuting Problems

5.1 Why global modal analysis fails

Let $H = \text{diag}(H_1, \dots, H_n)$ vary by location and let $G = G_0 \otimes I_d$ couple neighboring sites. In block form,

$$[H, G]_{ij} = (G_0)_{ij}(H_i - H_j),$$

so $HG \neq GH$ whenever coupled sites have different data blocks. The exact factor is then

$$R_{\text{true}}(\rho, \alpha) = \varrho(E_{\rho, \alpha}),$$

not a maximum of the scalar factors in (7).

5.2 Local freezing and patchwise commuting approximation

Let nonoverlapping patches $\{\Omega_r\}$ partition the spatial grid. On Ω_r , freeze the coefficient at a symmetric positive semidefinite representative block H_r . Let G_r be a symmetric patch regularizer with a specified closure, such as periodic or cosine, and define

$$\tilde{H}_r = I_{|\Omega_r|} \otimes H_r, \quad \tilde{G}_r = G_r \otimes I_d.$$

Then $\tilde{H}_r \tilde{G}_r = \tilde{G}_r \tilde{H}_r$. If $H_r z_j = h_{r,j} z_j$ and $G_r \phi_\xi = g_r(\xi) \phi_\xi$, the legitimate joint modes are $\phi_\xi \otimes z_j$ and their exact local factors are $e(h_{r,j}, g_r(\xi); \rho, \alpha)$.

5.3 Global minimax over local modes

One global pair is selected across all patches; patches do not receive separate ADMM penalties. With attained patch symbol endpoints g_r^- and g_r^+ , set

$$\mathcal{C} = \bigcup_{r,j} \{(h_{r,j}, g_r^-), (h_{r,j}, g_r^+)\}.$$

The patch-surrogate objective is exactly

$$U_{\text{patch}}(\rho, \alpha) = \max_{(h,g) \in \mathcal{C}} |e(h, g; \rho, \alpha)|. \quad (13)$$

The same finite branch argument produces a minimax pair over all local models and $\mathcal{I} \times (0, 2]$.

5.4 Optional variable-coefficient rate bounds

Rigorous noncommuting bounds may be evaluated after tuning. For example, if $\tilde{H} = \bigoplus_r \tilde{H}_r$ and $\tilde{G} = \bigoplus_r \tilde{G}_r$ are the full-size direct-sum surrogates on the patch partition, and

$$\delta_H = \|H - \tilde{H}\|_2, \quad \delta_G = \|G - \tilde{G}\|_2,$$

then the Cayley resolvent identity gives

$$R_{\text{true}}(\rho, \alpha) \leq U_{\text{patch}}(\rho, \alpha) + \frac{\alpha}{\rho} (\delta_H + \delta_G). \quad (14)$$

Each direct-sum block is simultaneously diagonalizable, so $\tilde{E}_{\rho, \alpha}$ is normal and $\|\tilde{E}_{\rho, \alpha}\|_2 = U_{\text{patch}}(\rho, \alpha)$ when the local symbol endpoints used in (13) are attained. Cross-patch regularizer couplings removed by the direct sum are included in δ_G . This bound requires no commutativity of the full operators but is conservative. It and the sharper block-Gershgorin, signed-rectangle, Perron-comparison, and truncated-kernel constructions are detailed in the supplementary information. None is used during deployed parameter selection.

6 Application to Diffeomorphic Image Registration

6.1 Linearized registration model

Let n grid points lie in dimension $d \in \{2, 3\}$, let Δx be the grid spacing, and let $q_i = \nabla I(x_i)$. At the current linearization, $v_i \in \mathbb{R}^d$ is the velocity increment at x_i and $I_t(x_i)$ is the scalar brightness residual. A damped linearized optical-flow subproblem can be written

$$\min_v \frac{\mu}{2} \sum_{i=1}^n (q_i^\top v_i + I_t(x_i))^2 + \frac{\nu}{2} \|v\|_2^2 + \frac{1}{2} v^\top G v, \quad (15)$$

where $\mu > 0$ weights data fidelity, $\nu \geq 0$ is an optional damping weight, and $G \succeq 0$ is the regularization Hessian. Setting $\nu = 0$ gives the undamped model. Splitting a copy $w = v$ places (15) in (1).

6.2 Rank-one data Hessian

The data Hessian is

$$H = \text{diag}(H_1, \dots, H_n), \quad H_i = \nu I_d + \mu q_i q_i^\top. \quad (16)$$

If $q_i \neq 0$, then H_i has eigenvalue ν with multiplicity $d - 1$ and eigenvalue $\nu + \mu \|q_i\|^2$ along q_i . If $q_i = 0$, all d eigenvalues equal ν . Therefore the attained global curvature endpoints are

$$h_- = \nu, \quad h_+ = \nu + \mu \max_i \|q_i\|^2. \quad (17)$$

6.3 Sobolev regularizer and symbol

The componentwise regularizer has

$$G = G_0 \otimes I_d.$$

For a first-order periodic Sobolev discretization with stiffness $\beta > 0$ and zeroth-order weight $\gamma \geq 0$,

$$g(\xi) = \gamma + \frac{4\beta}{(\Delta x)^2} \sum_{\ell=1}^d \sin^2 \left(\frac{\xi_\ell}{2} \right). \quad (18)$$

Thus $g_- = \gamma$. When every Nyquist-frequency corner is present, as on the even periodic grids used here, $g_+ = \gamma + 4\beta d/(\Delta x)^2$; otherwise the exact maximum over the discrete frequency set is used. Higher-order powers use the corresponding powered endpoints.

6.4 Registration-specific four-corner predictor

For a registration patch, choose a representative gradient $\bar{q}_r$ and set

$$H_r = \nu I_d + \mu \bar{q}_r \bar{q}_r^\top.$$

Local freezing at H_r recovers the separable tensor-product structure. The deployed four-corner surrogate uses the family of pixelwise coefficient freezes $H_r = H_i$; equivalently, the representative set includes every q_i . Hence h_+ in (17) is attained, and the rank-one endpoint collapse leaves the four branches

$$(h_-, g_-), \quad (h_-, g_+), \quad (h_+, g_-), \quad (h_+, g_+).$$

If only mean or median patch representatives are retained, these four branches remain a rectangular envelope but need not equal the patch spectrum. For these attained pixelwise endpoints, define $R_{4C} := U_{4C}$ and

$$(\rho_{4C}, \alpha_{4C}) \in \arg \min_{\rho \in \mathcal{I}, 0 < \alpha \leq 2} R_{4C}(\rho, \alpha).$$

Corollary 6.1 (Rank-one frozen-registration reduction). *Assume $d \geq 2$, the coefficient-frozen family contains every H_i , and the common regularizer spectrum attains g_- and g_+ . Also assume $\nu + \gamma > 0$, or remove the common null mode on an explicitly identified quotient subspace. Then the worst contraction factor over that family equals R_{4C} , and Theorem 4.3 returns a global minimizer of this frozen separable surrogate over $\mathcal{I} \times (0, 2]$. For the assembled variable-coefficient operator, the same pair is a predictor rather than a generally optimal pair.*

Remark 6.2. The four-corner reduction is exact for the separable commuting surrogate; the resulting parameter pair is therefore globally optimal over $\mathcal{I} \times (0, 2]$ for that surrogate, but serves as a predictor of the optimal parameters for the full variable-coefficient registration iteration.

In general, the selected pair need not belong to

$$\arg \min_{\rho \in \mathcal{I}, 0 < \alpha \leq 2} R_{\text{true}}(\rho, \alpha).$$

The predictor ignores global orientation interactions between spatially varying rank-one blocks; the experiments assess its practical transfer.

6.5 One-shot deployment policy

Algorithm 1: One-Shot Four-Corner oADMM Tuning *Inputs:* initial image gradients $\{q_i\}$, weights μ, ν , regularizer parameters and grid, and admissible penalty interval $\mathcal{I}$. *Output:* (ρ_{4C}, α_{4C}) .

1. From the initial image gradient, compute $h_- = \nu$ and $h_+ = \nu + \mu \max_i \|q_i\|^2$.
2. Compute g_- and g_+ analytically from the regularizer and grid.
3. Form the four endpoint pairs and their rational functions $\theta_c(\rho)$.
4. On the admissible interval, generate the finite algebraic penalty candidate set of Theorem 4.3.
5. At each candidate compute $a(\rho)$, $b(\rho)$, and $\alpha^*(\rho)$ (or an admissible default when $b(\rho) = 1$); select the pair minimizing R_{4C} .
6. Run inner oADMM with this fixed pair.
7. In one-shot deployment, reuse the pair at later outer registration iterations and pyramid levels unless an explicit retuning policy is enabled.

An *inner ADMM iteration* solves one quadratic problem with fixed H, G, b , so its optimal fixed parameters do not change during that solve. An *outer registration iteration* updates the deformation and relinearizes; the warped image gradients and H may then change. The one-shot policy reuses the initial pair after relinearization, thereby avoiding repeated parameter selection. The reported ablation tests per-pyramid-level retuning, not separate retuning at every outer Gauss–Newton iteration.

7 Experimental Protocol

The reference two-dimensional solver uses a three-level Gaussian pyramid, spatial image gradients, Gauss–Newton optical-flow linearization, Sherman–Morrison solves for the rank-one data blocks, and fast Fourier transform (FFT) solves for the Sobolev regularizer. Velocity increments are composed with the displacement. A backtracking line search requires objective decrease and a discrete Jacobian above 0.05. Pyramid transfer is audited and, if needed, its nontranslational component is halved until the same floor holds. All methods in a comparison share initialization, objective, stopping tolerances, line search, pyramid, and topology safeguards.

Three fixed baselines are distinguished throughout: canonical fixed ADMM uses $(\rho, \alpha) = (1, 1)$, fixed oADMM uses $(1, 1.8)$, and the external fixed comparator uses $(0.1, 1.0)$. The external comparator was chosen using separate validation images and was held fixed for both public benchmarks. Adaptive comparators are residual balancing and a safeguarded Barzilai–Borwein (BB) proxy; implementation details are given in the supplementary information.

The spectral tests use direct eigendecomposition only on tractable systems. The controlled spectral studies use $\mathcal{I} = [10^{-2}, 10^2]$; the real-image selector uses $\mathcal{I} = [10^{-4}, 10^3]$. Implementations use $\alpha \in [10^{-6}, 2]$ as a numerical realization of $0 < \alpha \leq 2$. The constant-coefficient test uses periodic $n \times n$ grids, $(h_-, h_+) = (0.2, 2.3)$, and first-order regularization with $(\beta, \gamma) = (0.2, 0.1)$. Controlled noncommuting tests separate periodic magnitude variation, smooth orientation variation, a sharp orientation interface, and random-gradient stress tests on $n \times n$ grids with $n = 3$ and $N = n^2$ sites. Writing $s_i = \sin(2\pi x_{i,1}/n)$ and $e_1 = (1, 0)^\top$, the first two constructions are

$$q_i^{\text{mag}} = 0.4(1 + as_i)e_1, \quad q_i^{\text{ori}} = 0.4 \begin{bmatrix} \cos(as_i) & \sin(as_i) \end{bmatrix}^\top,$$

with $a \in \{0.05, 0.2, 0.5\}$ and $a \in \{0.05, 0.2, 0.6\}$, respectively. The interface field has constant magnitude and a 90° orientation jump. For the random-gradient cases, a fixed periodic Gaussian white-noise field z is filtered with Gaussian scale $\sigma \in \{0.6, 1.0, 1.6\}$ pixels, differentiated, and rescaled:

$$q_i^{\text{rnd}} = c_\sigma \nabla(K_\sigma * z)_i, \quad \left(\frac{1}{N} \sum_i \|q_i^{\text{rnd}}\|^2 \right)^{1/2} = 0.4.$$

Thus these cases vary magnitude and orientation jointly. The Gaussian filter acts on the precursor field; differentiation and root-mean-square renormalization mean that σ is not a monotone measure of Hessian variation. Every controlled case uses $H_i = 0.05I_2 + q_i q_i^\top$ and the same periodic regularizer with $(\beta, \gamma) = (0.2, 0.1)$.

For each controlled case, define the normalized commutator

$$\chi = \frac{\|HG - GH\|_2}{\|H\|_2 \|G\|_2}.$$

For these tests, let R_{act} be the true spectral radius at the selected parameters, R_{ora} the radius at numerically oracle-optimized parameters, and U a rigorous enclosure evaluated at the selected pair. We report the spectral-radius regret

$$\mathcal{G}_{\text{ora}} := \frac{R_{\text{act}} - R_{\text{ora}}}{R_{\text{ora}}}$$

and the selected-point surrogate discrepancy

$$\mathcal{D}_{4C} := \frac{|R_{4C}(\rho_{4C}, \alpha_{4C}) - R_{\text{act}}|}{R_{\text{act}}}.$$

The first quantity measures contraction lost relative to the numerical oracle; the second measures agreement between the surrogate and true objectives only at the four-corner-selected pair. Certificate experiments additionally use the enclosure gap $(U - R_{\text{act}})/R_{\text{act}}$.

The real-image pilot uses a public-domain retinal image and an unrestricted immunohistochemistry image under held-out smooth known deformations. CIMA uses the five differently stained sections of the public lung-lesion-3 sample and all ten unordered pairs [3], resized to 256^2 after a common structural-channel preprocessing. FIRE uses all 134 landmarked pairs (14 A, 49 P, and 71 S) at 256^2 [8]. Landmarks are used only for evaluation. Timings use one central processing unit (CPU) thread for Basic Linear Algebra Subprograms (BLAS) and OpenMP. Reproducibility details, baseline safeguards, seeds, and certificate protocols are in the supplement.

8 Results

8.1 Exact four-corner validation

The first experiment verifies Theorem 4.2 under its exact assumptions. We construct constant-coefficient periodic quadratic systems on small $n \times n$ grids with prescribed data-curvature endpoints $(h_-, h_+) = (0.2, 2.3)$ and a first-order Sobolev regularizer with $(\beta, \gamma) = (0.2, 0.1)$. The operators commute, and their attained joint modal spectrum has the required Cartesian-product structure. These synthetic systems contain no registration images or landmarks.

The finite algebraic selector returned $(\rho, \alpha) = (0.543512, 1.898152)$ on every grid. Table 1 compares the four-corner value U_{4C} with the spectral radius of the full reduced matrix $E_{\rho, \alpha}$, obtained independently by direct eigendecomposition. The two agree to machine precision, verifying the predicted equality in this controlled setting. Figure 1(a) displays the same comparison.

Table 1: Exact four-corner verification on synthetic periodic systems.

n	Direct radius $\varrho(E)$	$ U_{4C} - \varrho(E) $
4	0.35313044	$< 10^{-15}$
6	0.35313044	$< 10^{-15}$
8	0.35313044	$< 10^{-15}$

8.2 Transfer under controlled noncommutativity

We next quantify predictor transfer after deliberately violating the theorem assumptions. Ten 3×3 periodic systems use spatially varying optical-flow Hessians with magnitude variation, smooth orientation variation, an abrupt orientation interface, or Gaussian-smoothed random gradient fields. The resulting H and G do not commute. At this tractable size, direct eigendecomposition gives the true spectral radius, and numerical minimization of that radius supplies oracle parameters.

Figure 1(b) relates predictor quality directly to noncommutativity. The magnitude-only and orientation-only cases had $\chi = 0.012$ – 0.139 and spectral-radius regret 0.003 – 0.087 . The abrupt interface had $(\chi, \mathcal{G}_{\text{ora}}) = (0.166, 0.193)$. The random-gradient stress tests varied magnitude and orientation simultaneously, produced the largest commutators ($\chi = 0.171$ – 0.239), and had regrets 0.238 – 0.296 . Across these ten deterministic cases, the descriptive Pearson correlation between χ and regret was 0.87 . This small controlled suite does not establish a universal relationship, but it shows that the observed high regrets coincide with stronger noncommutativity rather than with Gaussian smoothing itself.

As a separate pointwise diagnostic, the median selected-point discrepancy $\mathcal{D}_{4C}$ was 1.4×10^{-15} and its maximum was 2.8×10^{-3} ; this quantity is not plotted in Fig. 1(b). Pointwise agreement does not imply equal minimizers: the surrogate can match the true spectral radius at (ρ_{4C}, α_{4C}) while differing elsewhere in parameter space, where the numerical oracle finds a smaller radius. Detailed noncommuting certificates and scaling studies are reported in the supplementary information.

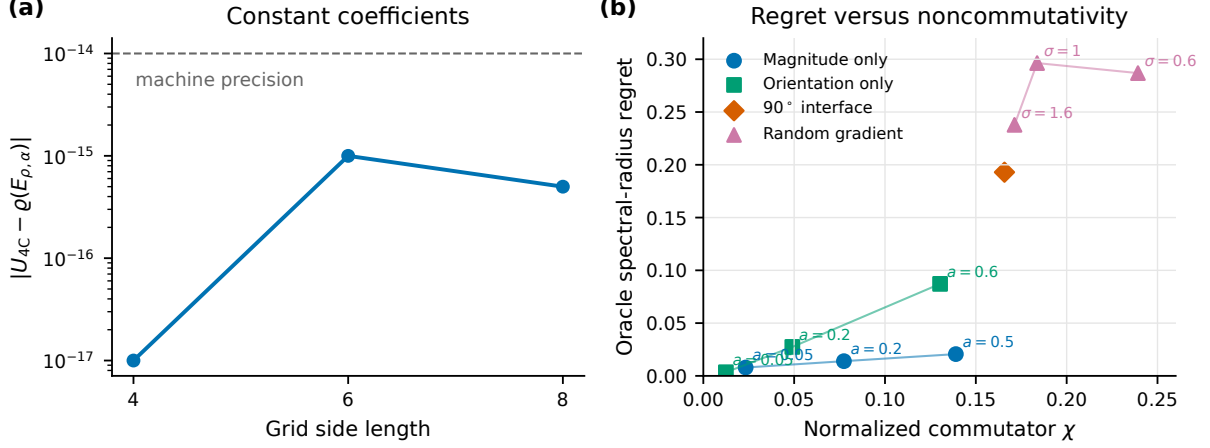


Figure 1: Validation of the four-corner reduction and predictor. **(a)** Constant-coefficient periodic systems satisfy the attained Cartesian-product assumptions; direct eigendecomposition of the full reduced matrix agrees with U_{4C} to machine precision. **(b)** Spatially varying optical-flow Hessians violate commutativity. The horizontal axis is the normalized commutator $\chi = \|HG - GH\|_2 / (\|H\|_2 \|G\|_2)$, and the vertical axis is the spectral-radius regret $(R_{\text{act}} - R_{\text{ora}})/R_{\text{ora}}$. Marker shapes identify magnitude-only, orientation-only, abrupt-interface, and random-gradient constructions; labels give the sinusoidal amplitude a or Gaussian precursor scale σ . Lines connect levels within a construction only as visual guides.

8.3 Real registration benchmarks

8.3.1 CIMA histology

The public lung-lesion-3 sample from the CIMA landmark repository [3] comprises five differently stained sections and all ten unordered pairs. Images supplied at 5% scale were converted to a common structural channel and resized to 256^2 ; landmarks supplied at 50% scale were rescaled for evaluation only. The problem is diffeomorphic registration of these pairs under the shared pyramid solver, comparing one-shot four-corner oADMM tuning with fixed-parameter, residual-balancing, and BB proxy baselines on target registration error (TRE), iterations, runtime, and discrete Jacobians. TRE is the Euclidean landmark error after registration.

The external fixed comparator $(\rho, \alpha) = (0.1, 1.0)$ was selected using separate validation images and then held fixed for CIMA and FIRE. Across five order-alternated repetitions of all ten pairs (50 paired runs), four-corner tuning reduced median total runtime by 20.1% (bootstrap 95% interval 19.4–20.7%), solver time by 23.5%, and iterations by 30.1%. Selection used 1.16% of proposed runtime. Median TRE was 7.33 pixels (2.02% of the image diagonal) and differed by less than 3×10^{-5} pixels from the fixed comparator. The median landmark-error improvement relative to the unregistered pairs was 36.5%. No global or interior nonpositive Jacobian was observed. One difficult stain pair retained TRE above 22 pixels, identifying initialization and cross-stain representation rather than inner ADMM convergence as the main failure mode.

Table 2: CIMA histology registration at 256^2 : medians of iterations and TRE over ten pairs, and of total time over 50 paired runs.

Method	Iterations	Time (s)	TRE (px)
Canonical fixed ADMM (1, 1)	1910.5	12.09	7.3310
Fixed oADMM (1, 1.8)	1129.5	7.37	7.3308
Residual balancing	1124.5	7.40	7.3310
BB proxy	435.5	3.44	7.3306
External fixed comparator (0.1, 1.0)	303.5	2.53	7.3306
Four-corner one-shot	210.0	2.02	7.3306

8.3.2 FIRE retinal registration

The public FIRE fundus registration benchmark [8]: all 134 landmarked pairs (14 A, 49 P, and 71 S), converted to a foreground-percentile-normalized green channel and resized to 256^2 . Landmarks are used only for evaluation. The problem is retinal diffeomorphic registration under identical phase-correlation initialization, pyramid, objective, and stopping criteria, with one CPU BLAS/OpenMP thread. One-shot four-corner tuning is compared with the external fixed comparator, residual balancing, a safeguarded BB proxy, and fixed oADMM on iterations, runtime, TRE, and topology.

Across all pairs, median iterations were 245 for one-shot four-corner tuning and 412 for the external fixed comparator, a paired median reduction of 39.6%. Median total times were 4.774 and 6.680 seconds, a 34.4% paired reduction (bootstrap 95% interval 30.7–35.7%); tuning accounted for 1.2% of proposed total time. The safeguarded BB proxy required 623.5 iterations and 9.435 seconds; the predictor was faster on 94.0% of pairs, with a median paired reduction of 51.7%. Median TRE differed from the fixed pair by -1.4×10^{-5} pixels, the maximum absolute difference was 3.9×10^{-4} pixels, and no method produced a nonpositive Jacobian.

Table 3: FIRE retinal registration: 134 pairs at 256^2 under identical initialization and solver protocol.

Method	Iter.	Time (s)	TRE (px)	TRE/diag. (%)
External fixed comparator (0.1, 1.0)	412.0	6.680	5.50178	1.51967
Residual balancing	1434.0	22.325	5.50211	1.51976
BB proxy	623.5	9.435	5.50175	1.51966
Fixed oADMM (1, 1.8)	1529.5	23.471	5.50194	1.51971
Four-corner one-shot	245.0	4.774	5.50177	1.51966

Pairwise percentage reductions need not equal percentage differences between separately reported method medians.

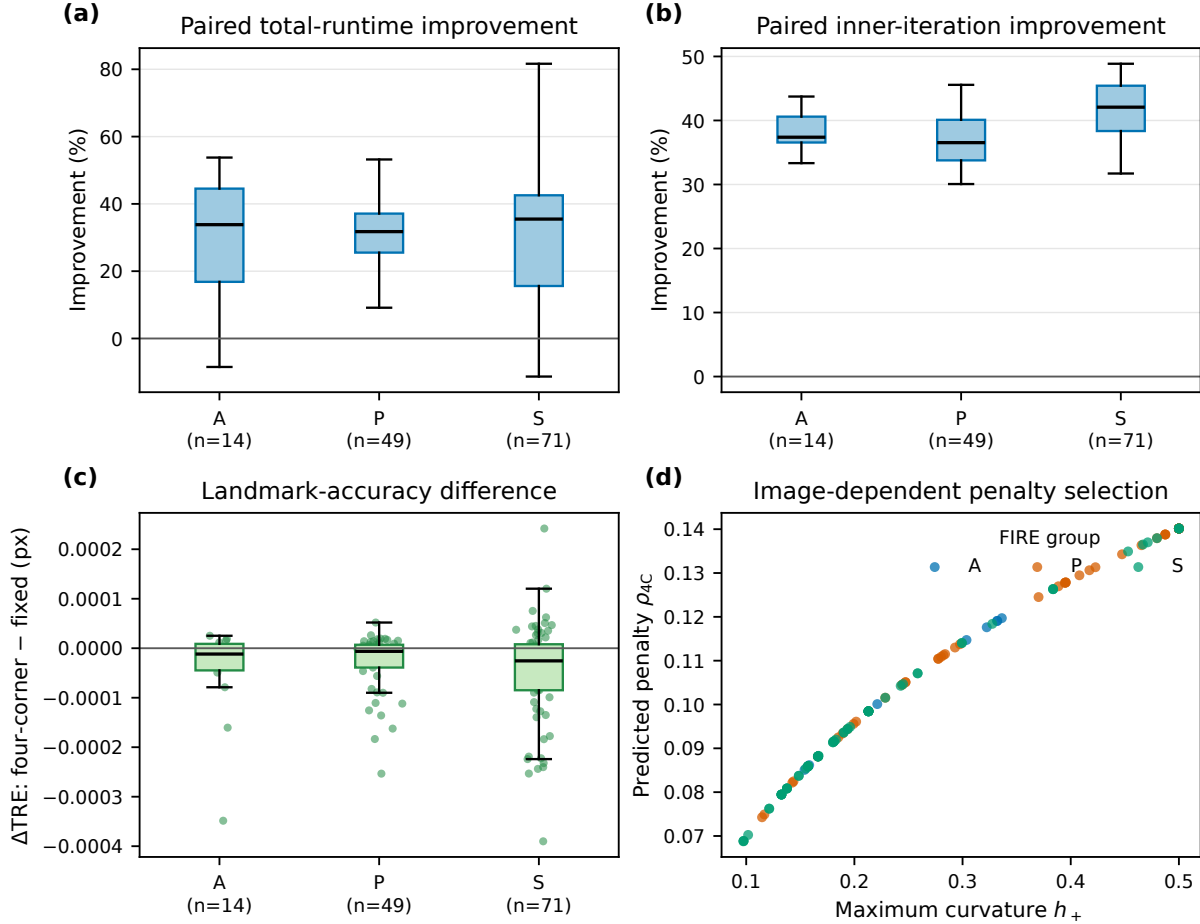


Figure 2: FIRE registration at 256×256 . (a) Pairwise total-runtime reduction and (b) inner-iteration reduction relative to the external fixed comparator. (c) Difference in TRE between four-corner and external fixed parameters. (d) Predicted penalty versus image-dependent maximum data curvature.

8.4 One-shot reuse ablation

On a balanced 18-pair FIRE subset, we compared one-shot reuse of the initial full-resolution parameter pair with retuning at every pyramid level. Retuning reduced median inner iterations from 243 to 222 but increased median total time from 2.651 to 3.895 seconds; median parameter-selection time increased from 0.030 to 0.110 seconds. Under this implementation, per-level retuning therefore traded fewer inner iterations for higher end-to-end runtime. The reported method uses one-shot reuse; retuning after every outer Gauss–Newton linearization was not evaluated.

9 Discussion

9.1 Why four curvature endpoints can suffice

Separability supplies independent data and regularizer modal coordinates. Monotonicity along each coordinate line then places the extreme modal values at endpoint pairs, and convexity of the relaxed magnitude preserves endpoint maximization. For registration, the rank-one data Hessian further reduces all data curvatures to ν and $\nu + \mu \max_i \|q_i\|^2$. The resulting four numbers encode the scale

balance between the data and regularization terms without reference to image dimension.

9.2 Transfer beyond the separable model

In the assembled registration operator, spatially varying data blocks interact with neighboring regularizer couplings. Smooth coefficient variation can retain locally meaningful curvature scales even though no global common modal basis exists. This provides a plausible explanation for the observed transfer of the surrogate-optimal pair, but not an exact characterization of R_{true} . The predictor omits global orientation interactions among the rank-one blocks.

9.3 Expected degradation and computational tradeoff

Rapid orientation changes, coefficient discontinuities, and strong noncommutativity increase the mismatch between frozen local models and the assembled operator. The larger regret in the abrupt-interface experiment is consistent with this mechanism. More accurate parameter selection can instead use repeated spectral calculations on the large variable-coefficient operator. The four-corner predictor occupies the complementary regime: it uses stronger structural assumptions to make tuning overhead negligible relative to the solve.

9.4 Limitations

The exact theorem requires an attained Cartesian-product joint modal spectrum; registration uses a noncommuting approximation. The experiments are two-dimensional and cover FIRE at 256^2 and one five-image CIMA tissue set at reduced resolution. The optional certificates are conservative, and the tested BB proxy represents one safeguarded adaptive implementation. Finally, the analysis concerns convex linearized subproblems and does not establish global convergence of the outer nonlinear registration method.

10 Conclusion

Quadratic oADMM with an attained Cartesian-product joint modal spectrum admits an exact four-corner characterization of its worst modal contraction. Analytic relaxation and a finite algebraic penalty candidate set then replace iterative spectral parameter search. For variable-coefficient registration, local freezing and the rank-one optical-flow Hessian turn this result into an inexpensive predictor. FIRE and CIMA experiments show reduced iterations and runtime with essentially unchanged landmark accuracy and observed topology. Further work may address tighter noncommuting analysis, volumetric registration, and related splitting methods.

Declarations

- **Funding.** No specific funding was received for this work.
- **Competing interests.** The authors declare no competing interests.
- **Ethics approval and consent to participate.** Not applicable; the study uses only publicly available, de-identified image datasets.
- **Consent for publication.** Not applicable.

- **Data availability.** FIRE [8] and CIMA [3] are publicly available from their original providers. The FIRE archive is not redistributed.
- **Materials availability.** Not applicable.
- **Code availability.** Source code, tests, configurations, processed JSON and CSV outputs, and preparation and evaluation scripts are available at <https://github.com/xhxuciedu/admm-registration>.
- **Author contributions.** Katherine Xie and Jie Wu conducted the research and wrote and edited the manuscript. Xiaohui Xie conceived the work and edited the manuscript.

A Proof of the Reduced Cayley Representation

Let (v^*, w^*, u^*) be a fixed point and write $\tilde{v}^k = v^k - v^*$, with analogous notation for w and u . Subtracting the fixed-point equations from (2)–(4) removes the affine term b . With

$$P_H = \rho(H + \rho I)^{-1}, \quad P_G = \rho(G + \rho I)^{-1},$$

the homogeneous v -update and relaxed variable are

$$\tilde{v}^{k+1} = P_H(\tilde{w}^k - \tilde{u}^k), \quad \hat{\tilde{v}}^{k+1} = [\alpha P_H + (1 - \alpha)I]\tilde{w}^k - \alpha P_H \tilde{u}^k.$$

Set

$$z^k = \hat{\tilde{v}}^{k+1} + \tilde{u}^k = [\alpha P_H + (1 - \alpha)I]\tilde{w}^k + (I - \alpha P_H)\tilde{u}^k.$$

The remaining updates are $\tilde{w}^{k+1} = P_G z^k$ and $\tilde{u}^{k+1} = (I - P_G)z^k$. Thus, for $s^k = [(\tilde{w}^k)^\top, (\tilde{u}^k)^\top]^\top$,

$$s^{k+1} = AB_\alpha s^k, \quad A = \begin{bmatrix} P_G \\ I - P_G \end{bmatrix}, \quad B_\alpha = \begin{bmatrix} \alpha P_H + (1 - \alpha)I & I - \alpha P_H \end{bmatrix}.$$

Here $A \in \mathbb{R}^{2N \times N}$ and $B_\alpha \in \mathbb{R}^{N \times 2N}$. Sylvester's determinant identity gives

$$\det(\lambda I_{2N} - AB_\alpha) = \lambda^N \det(\lambda I_N - B_\alpha A).$$

Hence the nonzero eigenvalues of AB_α and $B_\alpha A$ coincide, including algebraic multiplicity, and the $2N$ -dimensional state map has the N eigenvalues of $B_\alpha A$ together with N additional zeros.

Direct multiplication yields

$$B_\alpha A = (1 - \alpha)I + \alpha\{P_H P_G + (I - P_H)(I - P_G)\}.$$

Because $P_H = (I - C_H)/2$ and $P_G = (I - C_G)/2$,

$$P_H P_G + (I - P_H)(I - P_G) = \frac{1}{2}(I + C_H C_G).$$

Therefore

$$B_\alpha A = \left(1 - \frac{\alpha}{2}\right)I + \frac{\alpha}{2}C_H C_G = E_{\rho, \alpha},$$

which proves Proposition 3.1.

B Proof Details for the Four-Corner Theorem

Fix $\rho > 0$ and let $\mathcal{R} = [h_-, h_+] \times [g_-, g_+]$. The denominators in (9) are positive. For fixed g , the sign of $\partial\theta/\partial h$ is the sign of $g - \rho$ and is independent of h ; for fixed h , the sign of $\partial\theta/\partial g$ is the sign of $h - \rho$ and is independent of g . Consequently, replacing either coordinate of any point in $\mathcal{R}$ by a suitable endpoint cannot decrease θ , and a possibly different endpoint replacement cannot increase it. Applying this argument successively to both coordinates shows

$$\theta_{\min} := \min_{\mathcal{R}} \theta = \min_{c \in \text{corners}(\mathcal{R})} \theta_c, \quad \theta_{\max} := \max_{\mathcal{R}} \theta = \max_{c \in \text{corners}(\mathcal{R})} \theta_c.$$

Thus every attained modal value $t = \theta(h, g; \rho)$ lies in $[\theta_{\min}, \theta_{\max}]$.

For fixed $\alpha > 0$, the function $\phi(t) = |1 - \alpha + \alpha t|$ is convex. If $t = \lambda\theta_{\min} + (1 - \lambda)\theta_{\max}$ with $0 \leq \lambda \leq 1$, convexity gives

$$\phi(t) \leq \lambda\phi(\theta_{\min}) + (1 - \lambda)\phi(\theta_{\max}) \leq \max\{\phi(\theta_{\min}), \phi(\theta_{\max})\}.$$

Both interval endpoints are corner values, so U_{4C} bounds the magnitude of every modal eigenvalue. If H and G are simultaneously diagonalizable, $E_{\rho, \alpha}$ is diagonal in their common modal basis and its spectral radius is the largest modal magnitude. When all four endpoint pairs are attained common modes, whichever corner realizes the displayed maximum is an actual eigenmode, proving equality. Without endpoint attainment, the same argument proves only the rectangular upper envelope.

C Proof of Optimal Conditional Relaxation

Fix ρ and abbreviate $0 \leq a \leq b \leq 1$. The four-corner objective is

$$F(\alpha) = \max\{|1 - \alpha(1 - a)|, |1 - \alpha(1 - b)|\}, \quad 0 < \alpha \leq 2.$$

If $b = 1$, the second endpoint factor equals one for every α , while the first has magnitude at most one for $0 < \alpha \leq 2$ because $0 \leq a \leq 1$. Hence $F(\alpha) = 1$ throughout the admissible interval, and every relaxation is optimal.

Suppose now that $b < 1$. The unique unconstrained minimizer is obtained by equioscillation of the ordered endpoint factors:

$$1 - \alpha(1 - a) = -[1 - \alpha(1 - b)],$$

which gives $\alpha_0 = 2/(2 - a - b)$. This includes $a = b < 1$, for which both endpoint factors coincide and vanish at $\alpha_0 = 1/(1 - a)$. The function F is convex and is nonincreasing up to α_0 , so imposing $\alpha \leq 2$ gives $\alpha^* = \min\{2, \alpha_0\}$.

If $a + b \leq 1$, then $\alpha_0 \leq 2$, and substitution gives $F(\alpha^*) = (b - a)/(2 - a - b)$. If $a + b \geq 1$, then $\alpha_0 \geq 2$ and the constrained optimizer is $\alpha^* = 2$; because $2b - 1 \geq 1 - 2a$, its value is $F(2) = 2b - 1$. The formulas agree when $a + b = 1$.

D Proof of Finite Algebraic Selection

For a corner $c = (h_c, g_c)$, write

$$\theta_c(\rho) = \frac{N_c(\rho)}{D_c(\rho)}, \quad N_c(\rho) = \rho^2 + p_c, \quad D_c(\rho) = \rho^2 + s_c\rho + p_c,$$

where $s_c = h_c + g_c$ and $p_c = h_c g_c$. Because $\rho > 0$ and $h_c, g_c \geq 0$, every $D_c(\rho) = (\rho + h_c)(\rho + g_c)$ is strictly positive. Identical rational branches are deduplicated.

For distinct branches i and j , an intersection satisfies

$$N_i D_j - N_j D_i = \rho[(s_j - s_i)\rho^2 + p_i s_j - p_j s_i] = 0.$$

Thus its cleared polynomial has degree at most three (and, after removal of the inadmissible root $\rho = 0$, degree at most two). A relaxation-regime transition for a possible active pair satisfies

$$N_i D_j + N_j D_i - D_i D_j = 0,$$

a polynomial of degree at most four.

Let $\mathcal{Z}$ contain the endpoints of $\mathcal{I}$ and all roots in $\mathcal{I}$ of nonzero pairwise-intersection polynomials. On each connected component J of $\mathcal{I} \setminus \mathcal{Z}$, the strict ordering of all distinct branches is fixed. Hence there are fixed indices i and j such that $a = \theta_i$ and $b = \theta_j$ throughout J . Partition J further at the roots of $a + b = 1$. On each resulting open interval, the reduced objective is one of

$$F_{ij}^{(1)}(\rho) = \frac{\theta_j - \theta_i}{2 - \theta_i - \theta_j}, \quad F_j^{(2)}(\rho) = 2\theta_j - 1.$$

The denominator in the first regime is at least one and is therefore positive. When $b = 1$ in the second regime, the objective is the constant one and is handled by the piece endpoints.

After common denominators are cleared, $F_{ij}^{(1)} = P/Q$ with $\deg P \leq 3$ and $\deg Q \leq 4$. Its stationary equation $P'Q - PQ' = 0$ therefore has degree at most six. The stationary equation for $F_j^{(2)}$ has no larger degree, so six is a common upper bound. These degree counts also cover coincident curvature endpoints after identical branches have been removed.

The reduced objective is continuous on the compact interval $\mathcal{I}$, so it attains a global minimum. On the closure of each smooth piece, a nonconstant minimum lies at an endpoint or at an interior stationary point. If the objective is constant on a piece, either closure endpoint represents its value. Repeated roots do not create additional pieces. Identically zero intersection, transition, or stationarity polynomials are discarded; roots outside $\mathcal{I}$ and roots associated with inactive branches are also discarded. The union of the remaining endpoints, intersections, transitions, and stationary roots is finite and contains a global minimizer. Finally, evaluating the original objective at this finite superset removes roots made extraneous by denominator clearing and proves Theorem 4.3.

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